Armazéns de Dados

Departamento de Engenharia Informática (DEI/ISEP)
Paulo Oliveira
pjo@isep.ipp.pt

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Fátima Rodrigues (DEI/ISEP)

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Data Warehouse Optimization

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Indexing

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Indexes

- In almost all DWs, the **size of dimension tables is insignificant compared with the size of fact tables**
- **Second biggest thing in a DW** is the **size of indexes of the fact tables**
- Indexing mechanisms are used to **speed up access** to desired data
- Index file consists of records (called index entries) of the form:

search-key	pointer

- **Search key** is **an attribute** or **set of attributes** used to **look up records** in a file and **pointer points to the record** in the data table

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Index Types

- **B-Tree index**
  - Adequate for **high cardinality attributes**
  - May be built **on multiple columns**
  - Is the **default index type** for most databases, created on the primary key of a table
- **Bitmap index**
  - Adequate for **not high cardinality attributes** (*e.g.*: marital status)
  - Are a **major advance in indexing** that benefit DW applications
  - Are used both with **dimension tables and fact tables**, where there is **not high cardinality**
- **Others**
  - **Hash indexes** - array of *n* buckets or slots, each one containing a pointer to a row.
  - **Star indexes** - Some DBMSs use **additional index structures or optimizations strategies adequate to the n-way join** problem, inherent to a star query (*e.g.*: Red Brick).

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Bitmap Index

- Bitmap is simply an **array of bits**
- Bitmap index on an attribute has a **bit for each value of the attribute**
  - Bitmap has as many bits as distinct values
  - In a Bitmap for value v, the bit for a record is 1 if the record has the value v for the attribute, and is 0 otherwise

Id_Client	Gender	City	Income Level
145023	M	Brooklin	L1
145025	F	Jonestown	L2
154265	F	Perryridge	L4
265453	M	Brooklin	L1
645654	F	Perryridge	L3

Bitmap Index Structure

- Bitmap index for *Color* and *Type*

Cars			
ID	Type	Color	..other..
1DGS902	Sedan	White	...
1HUE039	Sedan	Silver	...
2UUE384	Coupe	Red	...
2ZUD923	Coupe	White	...
3ABD038	Sedan	Silver	...
3KES734	Coupe	White	...
3IEK299	Sedan	Red	...
3JSU823	Sedan	Silver	...
3LOP929	Coupe	Silver	...
3LMN347	Coupe	Red	...
3SDF293	Sedan	White	...

Color Bit Map Index		
Silver	Red	White
0	0	1
1	0	0
0	1	0
0	0	1
1	0	0
0	0	1
0	1	0
1	0	0
1	0	0
0	1	0
0	0	1

Type Bit Map Index	
Sedan	Coupe
1	0
1	0
0	1
0	1
1	0
0	1
1	0
1	0
0	1
0	1
1	0

- Bitmap index **often requires less storage space** than a conventional B-tree index
- For an attribute with **many distinct values**, can **exceed the storage space** of a conventional B-tree index

Using Bitmap Indexes

- Bitmap indexes **are useful for queries on multiple attributes** but not particularly useful for single attribute queries
- Queries are answered using **bitmap operations**
  - Intersection (AND)
  - Union (OR)
  - Negation (NOT)
- Each operation takes a bitmap vector and **applies the operation** to get the result bitmap vector
- Bit manipulation and searching is so fast, that the speed of **query processing with a bitmap index can be 10 times faster** than with a conventional B-tree index
- Databases support: DB2; Informix; Ingres; Oracle; PostgreSQL

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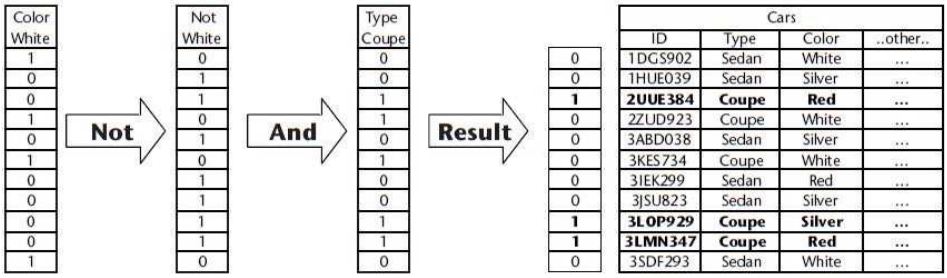
Queries on Bitmap Indexes

- Example of a query: find all cars that are not white and which are coupes



- Database is able to resolve the query using the **bitmap vectors** and **Boolean operations**
- It does not need to **touch the data** until it has **isolated the rows that answer the query**

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Indexing Dimension Tables

- Dimension tables have a **single column primary key** – must have **one unique index on that key**
- Small dimension tables seldom benefit from additional indexing
- Large dimension tables (*e.g.*: customer, product)
  - **Single-column bitmap or B-tree indexes** on **dimension attributes** that are most commonly used for:
    - **applying filters**
    - **grouping**
    - **used in a join condition**

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Indexing Fact Tables

- Fact table index must be a **B-Tree index** on the **primary key**
- Primary key, and the primary key index, must have **date-key in the first position** in the primary key index
  - Incremental loads are keyed by date
  - Most DW queries are constrained by date
- Create a **single-column index on each fact table key** and let the optimizer combine those indexes as appropriate to answer the queries
- If many queries constrain fact measures (amount, quantity) they must also be included in indexes – **non-key fact table indexes** are **single-column indexes**

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Partitions

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Partitions

- Segments tables in **small tables** for administrative purposes and to **improve performance**
  - SQL instructions manipulate the partitions instead of all the table
- Partitions are particularly useful
  - When they are defined in **different disks**
  - To manage **big fact tables**, **large dimensions**, and **their indexes**
- **Index Partitions**
  - **Global Index** – indexes all rows, regardless of the partition
  - **Local Index** – indexes only rows in a specific partition
- DBMSs support **different combinations** of table and index partitions
  - Partitioned table with index partitioned
  - Partitioned table with index not partitioned
  - Not partitioned table with index partitioned

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Horizontal Partitions

- Breaks a table by **placing different rows into different physical files** based on a condition
- Makes sense when **different categories of rows** of a table **are processed separately**
- **Best way to horizontally partition tables is by date**, with data segmented by year, semester, or quarter into separate storage partitions
  - ➥Date is often a qualifier in queries so, the **needed partitions are quickly found**
- Each file created from horizontal partitioning **has the same structure**

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Vertical Partitions

- **Distributes the columns of a table into separate physical records**, repeating the primary key in each of the records
- **Schema is different** from partition to partition
- **Less used** than horizontal partitions
- Reasons to use:
  - **Performance** – Updates and queries perform better because the database is able to buffer more rows at a time
  - **Change history** – some attributes change more frequently
  - **Large text** – placing large text columns in their own tables

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Methods of Horizontal Partitions

- **Range partitioning**
  - Partitions are created according to a **range of values for an attribute or a set of attributes**
  - Assumes that the **values** of the attributes used in the partition respect an **order**
- Best results occur when:
  - Size of partitions are **uniform**
  - Access to data is distributed through a **small number of partitions**

```
CREATE TABLE sales_range
(sales_id NUMBER(5),
sales_name VARCHAR(30),
sales_amount NUMBER(10),
sales_date DATE)
PARTITION BY RANGE (sales_date)
(PARTITION sales_2019 VALUES LESS THAN(TO_DATE('01/01/2020','DD/MM/YYYY')) tablespace ts1,
 PARTITION sales_2020 VALUES LESS THAN(TO_DATE('01/01/2021','DD/MM/YYYY')) tablespace ts2,
 PARTITION sales_2021 VALUES LESS THAN(TO_DATE('01/01/2022','DD/MM/YYYY')) tablespace ts3,
 PARTITION sales_2022 VALUES LESS THAN(TO_DATE('01/01/2023','DD/MM/YYYY')) tablespace ts4);
```


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Methods of Horizontal Partitions

- **List partitioning**
  - Partitions are created according to a **list of values of an attribute** explicitly specified
  - Useful when **data is not ordered nor has a relation**
  - **Values** of the attribute used in the partition **must be previously known**

```
CREATE TABLE sales_list
(sales_id NUMBER(5),
 sales_name VARCHAR(30),
 sales_region VARCHAR(20),
 sales_amount NUMBER(10),
 sales_date DATE)
PARTITION BY LIST(sales_region)
( PARTITION sales_north VALUES('Porto', 'Braga', 'Vila Real') tablespace ts1,
  PARTITION sales_central VALUES ('Lisboa', 'Santarém', 'Setúbal') tablespace
   ts2,
  PARTITION sales_south VALUES('Beja', 'Faro') tablespace ts3,
  PARTITION sales_other VALUES(DEFAULT) tablespace ts4);
```


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Partitions in Data Warehouses

- Table partitions can **dramatically improve query performance on large fact tables**
  - Performing a query that **requires a year of data from a partitioning fact table** that has ten years of data, can go **directly to the partition that contains the data** without scanning all the table
- Common practice: **partitioning the fact table by date** (e.g., year, semester, quarter)
- **Partitions are maintained** by the DB administrator or by the DW administrator

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Aggregates

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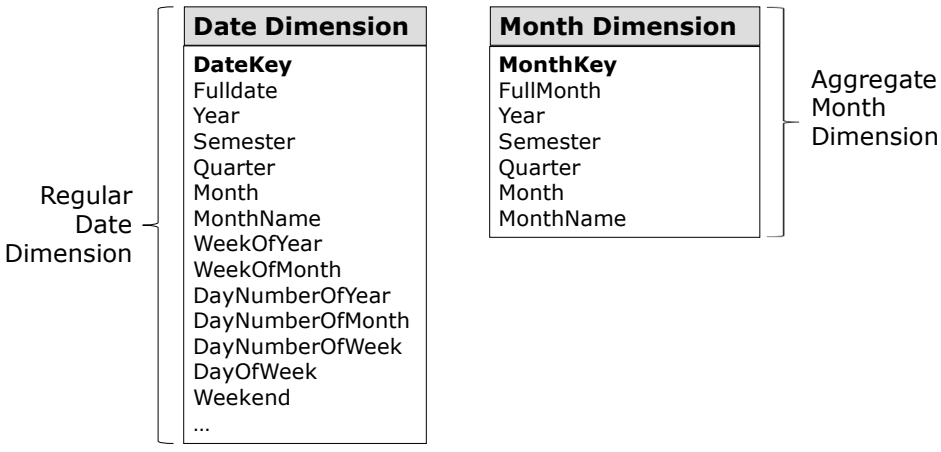
Aggregates

- **Summarization of a set of measures**, stored into fact tables with the purpose of **accelerating queries**
- Always associated with **one or more dimensions** that are **aggregated or not**
  - Example: **Sales** by ***product category***, by ***store*** , by ***date***

Fact aggregate dimension dimensions

- Can have a very significant **effect on performance**
  - Speeding queries by a factor that goes from 100 to 1000
  - No other means exist to provide such performance gain

Regular vs Aggregate Dimension



Aggregates Advantages/Disadvantages

- **Advantages**
  - Improve performance of the DW
  - Transparent to end-users and applications
  - Shared by many users
- **Disadvantages**
  - Only speed up the answers for the questions previously known, i.e., previously calculated and stored
  - Need constant attention by the DW administrator to:
    - Build new aggregates adequate to the more frequent questions made by the users
    - Eliminate aggregates that are not useful
  - Spend disk space

Deciding What to Aggregate

- Common Business Users' Requests
  - Reviews should be performed to determine **which attributes are commonly used for grouping**, considering:
    - Each attribute individually
    - Combinations of attributes
      - Within a dimension
      - Among dimensions
  - Not all attributes or combinations of attributes **are used together**

Deciding What to Aggregate

- **Statistical Distribution of Data**
  - **Number of attribute values** that are **candidates for aggregation**
    - 1.000.000 products exist in the product dimension
      - Aggregates at next level 500.000 – would not provide a significant improvement
      - Level that aggregates to 75.000 would be a strong pre-stored aggregate
    - Date dimension (5 years)
      - **Day** **1826**
      - **Month** **60**
      - **Quarter** **20**
      - **Year** **5**

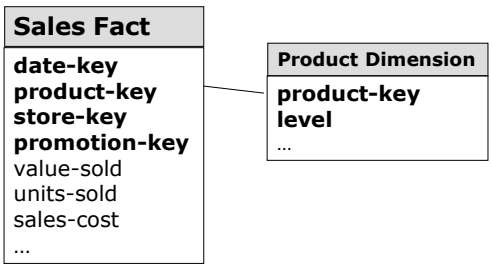
      Product dimension

SKU	**2023**
Product	**723**
Brand	**44**
Category	**15**

      - Month aggregate cuts data to 1/30 of detail size
      - Brand aggregate cuts data to about 1/50 of the detail size

Techniques to Store Aggregates

1. Aggregate facts and aggregate dimensions are stored in **new tables, separated from the base atomic data**
2. Aggregate facts are stored in the atomic fact table, and **level attributes** are stored in the dimensions
   - *Level* attribute show the aggregation level of each row



- Original rows are filled with *level* = 'Base'
- Aggregates for *Category* are filled with the *level* = '*Category*'

Comparing the Two Methods

- Number of records created **is the same** in both methods
- **Separated Tables**
  - Tables that correspond to the aggregates are not visible to end-users
  - Aggregates in separated tables can be easily created, deleted, loaded and indexed
- **Level Attributes**
  - Can conduct a **double-count additive facts totals** – all the queries must restrict the level attribute – if not, all the values are included/added
  - **Wasting disk storage** – adding the aggregates in the base fact table implies to increase the attribute width for all the records
  - **Dimension tables are more complicated** – for the records corresponding to the aggregates, many attributes are filled with '*not applicable*' or with *null*